

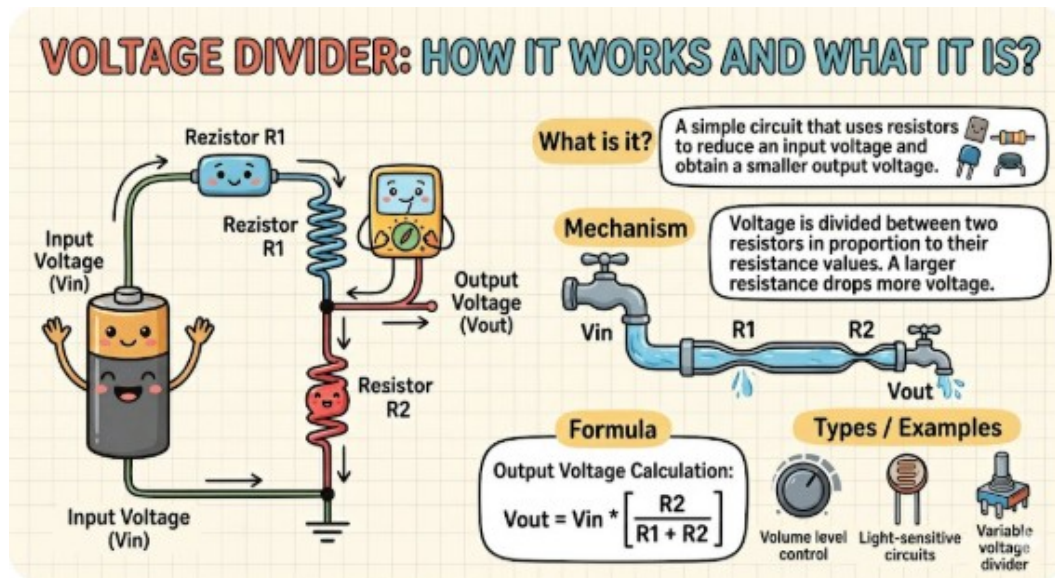
Physics Guide: COMPREHENSİVE TECHNİCAL REPORT: Voltage Divider

Classification: TTECHNICAL GUIDE/ EDUCATIONAL

DOCUMENTARY DATE: APRIL 2026

Xumar Bağiyeva, Aysel İsmayılova, Rəşadə Zülfüqarlı, Ramil Musazadə

Verified by: Physics Teacher Azerbaijan Telman Askeraliyev (Fizika Muellimi, Azerbaijan, Baku)



Voltage Divider: Technical Overview

1. Definition

A **Voltage Divider** is a simple linear circuit that produces an output voltage (V_{out}) that is a fraction of its input voltage (V_{in}). It is one of the most fundamental building blocks in electrical engineering, used to create a specific reference voltage or to attenuate signals.

2. The Operating Mechanism

The mechanism is based on **Ohm's Law** and **Kirchhoff's Voltage Law**. When two resistors are connected in series, the total input voltage is distributed across them proportional to their resistance values.

- The current (I) flowing through both resistors is the same.
- The voltage "drops" as it passes through each resistor.

- The output is typically measured across the second resistor (R_2).
-

3. Mathematical Formula

To determine the exact output voltage, the following formula is applied:

$$V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$$

- V_{in} : Input source voltage.
 - R_1 : The first resistor (connected to the source).
 - R_2 : The second resistor (connected to ground).
 - V_{out} : The reduced voltage available for the load.
-

4. Key Applications

Voltage dividers are not usually used for high-power applications (because they waste energy as heat), but they are essential for:

- **Level Shifting:** Converting a 5V signal to a 3.3V signal for microcontrollers.
 - **Sensor Interfacing:** Using LDRs (Light Dependent Resistors) or Thermistors to measure environmental changes.
 - **Potentiometers:** Acting as a variable voltage divider to control volume or brightness.
 - **Battery Monitoring:** Reducing high battery voltages to a range that an Analog-to-Digital Converter (ADC) can safely read.
-

5. Limitations and Considerations

- **Loading Effect:** If you connect a heavy "load" to the output, it acts as a parallel resistor to R_2 , which changes the output voltage.
- **Power Dissipation:** Resistors convert excess voltage into heat. High-current circuits require more efficient methods like "Buck Converters" instead of simple dividers.

9. Fun Fact: The "Human" Voltage Divider

Did you know that a **Volume Knob** on a guitar or an old radio is actually a "sliding" voltage divider? As you turn the knob, you are physically moving the V_{out} pin closer to ground (lower volume) or closer to the input signal (higher volume). It's the most tactile way we interact with this circuit every day!
